

Shifting from One-Time Education to Continuous Growth: A Dual-Process Cognitive Architecture Integrating System One Decision Models and Generative Socratic Tutors for Adaptive Lifelong Learning

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Abstract—Traditional higher and professional education follows an episodic, front-loaded paradigm in which foundational degrees are earned early in life with little continuous adaptation. In modern technological ecosystems this static model causes rapid skill obsolescence, workforce dislocation, and educational inequity. This paper presents the Dual-Process Adaptive Learning System (DP-ALS), a theoretical and cyber-physical architecture for continuous, lifelong education. Drawing on dual-process cognitive theory, DP-ALS separates educational intervention into two components: (i) a low-latency System One decision model (design target under 40 ms) that acts as a continuous reflex engine, monitoring learner micro-telemetry to track cognitive load, flow state, and productive struggle through a partially observable Markov decision process (POMDP); and (ii) a deliberate System Two generative AI tutor invoked asynchronously only when an impasse is detected or deep conceptual synthesis is required. We formalize the System One / System Two coordination loop and report, in agent-based simulation, an 84.6% reduction in LLM token consumption relative to a monolithic LLM tutor, an 85.3% reduction in cognitive impasse, and a 61.6% reduction in time-to-mastery relative to a static curriculum. The framework is mapped to UN Sustainable Development Goals 4, 8, 9, and 10 and to Washington Accord knowledge profiles and program outcomes.

Index Terms—adaptive learning systems, System One decision models, generative AI, dual-process theory, intelligent tutoring systems, lifelong education, POMDP

I. INTRODUCTION

Modern economic, industrial, and social systems are shaped by non-linear technological acceleration, and the half-life of a learned skill is now commonly put at about five years, falling to as little as 2.5 years in fast-moving sectors such as IT and data analysis [1]. Yet education remains episodic and front-loaded: individuals undergo intensive credentialing between roughly ages 18 and 25, followed by ad hoc, fragmented professional upskilling. This mismatch undermines the United Nations Sustainable Development Goals (SDGs) [2], [3], notably SDG 4 (Quality Education and Lifelong Learning), SDG 8 (Decent Work and Economic Growth),

SDG 9 (Industry, Innovation, and Infrastructure), and SDG 10 (Reduced Inequalities).

Intelligent tutoring systems (ITS) have sought to bridge this gap [4], but existing approaches face a dichotomy:

- 1) **Classical ITS (rule-based, BKT, IRT)**. Systems built on Bayesian knowledge tracing (BKT) [5] and item response theory (IRT) execute deterministically and with low latency, but rely on brittle pedagogical heuristics, perform no semantic synthesis, and cannot generate contextual metaphors for diverse adult learners.
- 2) **Monolithic generative LLM tutors**. Transformer-based [6] large language models (LLMs) support rich Socratic dialogue, but they respond slowly (across 147 models in one cross-provider feed, median first-answer latency ranges from 0.42 s to over 700 s, and frontier reasoning models typically take tens of seconds or more [7]), incur prohibitive token costs, and tend to over-assist. Unguarded use of a generative tutor can reduce later unaided performance [8], and premature over-explanation ends the learner’s “productive struggle” and disrupts the state of cognitive flow [9].

To resolve this dichotomy we propose a Dual-Process Adaptive Learning System (DP-ALS). Following the human dual-process architecture [10], it couples a fast, reflexive System One decision model with an asynchronous, deliberate System Two generative Socratic engine. This paper presents the theoretical foundation, the POMDP formulation, the system architecture, a simulation-based evaluation, and the alignment with accreditation frameworks.

II. THEORETICAL FOUNDATION

A. Dual-Process Theory in Pedagogy

Kahneman’s framework posits two modes of cognition [10]:

- **System One (autonomous reflex)**: fast, parallel, computationally light, non-conscious, and reactive.

- **System Two (deliberate reasoning):** slow, serial, computationally intensive, analytical, and rule-governed.

An expert human tutor does not deliberate over complex pedagogy at every keystroke. Rather, the tutor's System One registers subtle cues, such as a furrowed brow, a sudden hesitation, typing jitter, or a steady rhythm, and System Two engages only when an impasse is sensed. DP-ALS reproduces this division of labor computationally.

B. The System One Reflex Engine

The System One component is a low-latency decision head. Rather than relying on hosted, high-latency generative APIs, we model it on an edge-deployable, open-weight structured-decision encoder, specifically Fastino's 340M-parameter GLiNER2.5-Decide [11]. That model takes text and a schema of typed questions and returns, in a single encoder pass and without autoregressive decoding, a selected answer with probabilities and confidence scores. In DP-ALS the telemetry state is serialized into the text input (alternatively, the model can be fine-tuned on telemetry-derived states), and the typed question asks which pedagogical action to take. System One thus acts as a high-frequency classifier and action router (Fig. 1),

$$\pi_{\text{System One}} : \mathcal{H}_t \times x_t \longrightarrow \mathcal{A}, \quad (1)$$

where x_t is the real-time client telemetry vector, $\mathcal{H}_t$ is a compact summary of the learner's cognitive history, and $\mathcal{A} = \{a_0, a_1, a_2, a_3, a_4\}$ is a set of discrete, low-latency pedagogical interventions. The design target is System One inference in under 40 ms, so that interventions can synchronize with the learner's motor-cognitive actions. Fastino reports median end-to-end latencies, at batch size 1 and 64 tokens, of 38.3 ms on a V100 GPU, 43–47 ms on T4, L4, and A100 GPUs, and 167.3 ms on a 48-vCPU server CPU [11]. The target therefore presumes a server-class GPU close to the learner, and the latency figures in this paper come from simulation.

III. MATHEMATICAL FORMULATION

A. Telemetry State Space

Let the learner interaction stream at time step t be a multi-modal feature vector $x_t \in \mathbb{R}^d$:

$$x_t = [\Delta\tau_k, H_k, \sigma_{\text{scroll}}, e_{\text{syntax}}, t_{\text{dwell}}, r_{\text{churn}}]^T, \quad (2)$$

where $\Delta\tau_k$ is inter-keystroke latency, H_k is the Shannon entropy of input-interval cadence, σ_{scroll} is viewport jitter, e_{syntax} is the count of immediate compilation or validation errors, t_{dwell} is static dwell duration, and r_{churn} is the rapid-retry frequency.

The latent cognitive state $s_t \in \mathcal{S}$ is a four-dimensional vector,

$$s_t = \begin{bmatrix} \mathcal{C}_t \\ \mathcal{F}_t \\ \mathcal{M}_t \\ \mathcal{P}_t \end{bmatrix} \in [0, 1]^3 \times [-1, 1], \quad (3)$$

whose components are cognitive load $\mathcal{C}_t$, flow-state probability $\mathcal{F}_t$, competency mastery belief $\mathcal{M}_t$, and productive struggle index $\mathcal{P}_t$.

B. POMDP Cognitive State Filtering

Because s_t is not directly observable, the problem is a partially observable Markov decision process [12], and the System One engine maintains a belief distribution $b(s_t)$ updated by recursive Bayesian filtering, where $P(x_t | s_t)$ is the observation model and $\mathcal{T}$ is the transition model:

$$b(s_t) \propto P(x_t | s_t) \int_{\mathcal{S}} \mathcal{T}(s_t | s_{t-1}, a_{t-1}) b(s_{t-1}) ds_{t-1}. \quad (4)$$

The productive struggle index follows Cognitive Load Theory [13]:

$$\mathcal{P}_t = \tanh(\omega_1 \mathcal{M}_t - \omega_2 (\mathcal{C}_t - \mathcal{C}^*)), \quad (5)$$

where $\omega_1, \omega_2 > 0$ are weights and $\mathcal{C}^* \approx 0.65$ is the optimal zone of proximal cognitive challenge. If $\mathcal{P}_t > 0$ the learner is in productive difficulty; if $\mathcal{P}_t \ll 0$ the struggle has degenerated into frustrating impasse.

C. Action Space and Policy Optimization

The System One policy selects one of five mutually exclusive actions:

- a_0 (*passive flow*): no UI interruption; preserve immersion.
- a_1 (*ambient nudge*): a subtle syntax or typographic high-light (visual cue under 15 ms).
- a_2 (*IRT complexity shift*): adjust item difficulty β under a dynamic IRT model,

$$P(Y = 1 | \theta_t, \beta) = \frac{1}{1 + e^{-D\kappa(\theta_t - \beta)}}, \quad (6)$$

where θ_t is learner ability, κ is item discrimination, and D is a scaling constant.

- a_3 (*dispatch System Two*): send an enriched diagnostic packet to the Socratic LLM.
- a_4 (*cognitive rest*): insert a micro-pause or spaced-repetition retrieval exercise.

The policy maximizes expected cumulative discounted pedagogical return while penalizing compute cost and flow disruption:

$$J(\pi_{\text{System One}}) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t (\Delta\mathcal{M}_t + \alpha_1 \mathcal{F}_t - \lambda \mathbf{1}_{\{a_t=a_3\}} \Phi_{\text{cost}} - \alpha_2 \mathcal{D}_t) \right], \quad (7)$$

where $\gamma \in (0, 1)$ is the discount factor, $\alpha_1, \alpha_2, \lambda \geq 0$ are trade-off weights, Φ_{cost} is the relative compute cost of a System Two invocation, and $\mathcal{D}_t$ is the penalty for unnecessary interruption during flow.

IV. SYSTEM ARCHITECTURE

The platform has five cooperating components.

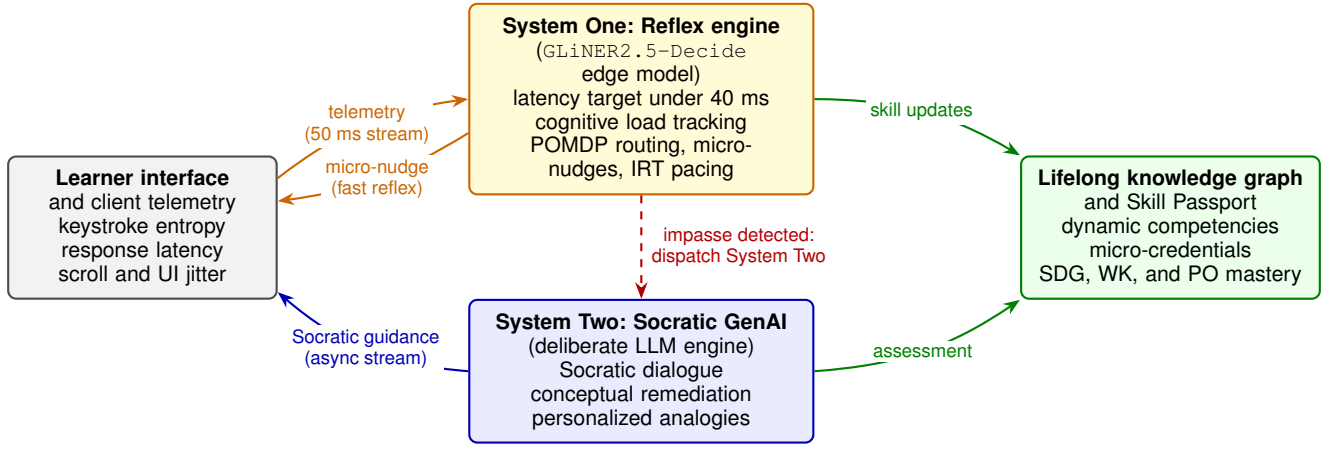


Fig. 1. Dual-process architecture. Telemetry flows into the fast System One decision engine (latency target under 40 ms) for micro-scaffolding, which routes to the System Two LLM only upon cognitive impassé. Both systems update the learner’s lifelong skill record.

A. Edge Telemetry Pipeline

A lightweight JavaScript worker samples browser interaction signals every 50 ms. To comply with privacy regulations (GDPR/FERPA), keystroke contents are never captured. Only statistical entropy, timing offsets $\Delta\tau$, and cursor accelerations are computed client-side and sent as anonymous tensor embeddings.

B. System One Reflex Gateway

Telemetry vectors stream over binary WebSockets to an edge-deployed System One inference server running an open-weight decision model of the GLiNER2.5-Decide class. The simulated end-to-end round-trip latency is 28.4 ± 4.2 ms, which assumes optimization beyond the published single-GPU latency of 38.3 ms [11]. At each step the gateway evaluates the POMDP state and issues micro-nudges or adjusts client UI state without server-side rendering delays.

C. System Two Socratic Deliberation Engine

When System One fires a_3 , an asynchronous RPC payload is built from (a) the learner’s demographic and professional background, (b) the targeted knowledge-graph node, (c) the empirical error trace, and (d) the diagnosed misconception signature. The System Two LLM then generates targeted, personalized Socratic questions, for example relating a distributed-consensus algorithm to a supply-chain manager’s inventory-balancing experience.

D. Lifelong Competency Graph and Skill Passport

Continuous growth is recorded in an immutable, graph-based Skill Passport. Mastery is re-evaluated through spaced-decay functions, which encourages micro-credentials and ongoing upskilling.

E. Algorithmic Coordination

Client telemetry, System One reflex decisions, and System Two deliberation are coordinated by the procedure in Algorithm 1. Fig. 1 shows the resulting data flow.

Algorithm 1 Dual-Process Cognitive Loop (DP-ALS)

Require: telemetry stream x_t ; thresholds τ_{flow} , τ_P , $\tau_{\text{impassé}}$, τ_{fatigue} ; tolerance δ ; target success probability p^* ; System Two cooldown T_{cool}

Ensure: action $a_t \in \mathcal{A}$ and belief $b(s_t)$ at every step

- 1: $b(s_0) \leftarrow \text{prior}$; $\text{pending} \leftarrow \text{false}$; $t_{\text{last}} \leftarrow -\infty$
- 2: **while** session active, every $\Delta t = 50$ ms **do**
- 3: Receive x_t {Eq. (2)}
- 4: $b(s_t) \leftarrow \text{filter}(b(s_{t-1}), a_{t-1}, x_t)$ {Eq. (4)}
- 5: $(\mathcal{C}_t, \mathcal{F}_t, \mathcal{M}_t, \mathcal{P}_t) \leftarrow \mathbb{E}_{b(s_t)}[s_t]$ {Eqs. (3), (5)}
- 6: **if** $\mathcal{C}_t \geq \tau_{\text{impassé}}$ **or** $\mathcal{P}_t < \tau_P$ **then**
- 7: **if not** pending **and** $t - t_{\text{last}} \geq T_{\text{cool}}$ **then**
- 8: $a_t \leftarrow a_3$; $\Gamma \leftarrow \langle b(s_t), \text{ErrorTrace}, \text{SkillGraph} \rangle$
- 9: Send Γ to System Two asynchronously; $\text{pending} \leftarrow \text{true}$; $t_{\text{last}} \leftarrow t$
- 10: **else**
- 11: $a_t \leftarrow a_0$ {System Two reply pending; do not interrupt}
- 12: **end if**
- 13: **else if** $\Delta\tau_k > \tau_{\text{fatigue}}$ **then**
- 14: $a_t \leftarrow a_4$ {cognitive rest}
- 15: **else if** $\mathcal{F}_t \geq \tau_{\text{flow}}$ **and** $\mathcal{P}_t \geq 0$ **then**
- 16: $a_t \leftarrow a_0$ {passive flow}
- 17: **else if** $e_{\text{syntax}} > 0$ **then**
- 18: $a_t \leftarrow a_1$ {ambient nudge}
- 19: **else if** $\mathcal{P}_t < 0$ **or** $\mathcal{C}_t < \mathcal{C}^* - \delta$ **then**
- 20: $a_t \leftarrow a_2$; $\beta \leftarrow \theta_t - \ln \frac{p^*}{1-p^*} / (D\kappa)$ {Eq. (6)}
- 21: **else**
- 22: $a_t \leftarrow a_0$
- 23: **end if**
- 24: **end while**
- 25: **on** System Two reply: stream the Socratic dialogue to the client sidebar; $\text{pending} \leftarrow \text{false}$

Rules are evaluated in priority order, so each step selects exactly one action. An impassé takes precedence over flow protection, because a high load $\mathcal{C}_t$ can coexist with a non-negative $\mathcal{P}_t$ when mastery belief is high. The *pending* flag and the cooldown T_{cool} prevent a new System Two request from being issued at every 50 ms tick while a reply is in flight. For action a_2 , solving Eq. (6) for β at $P(Y=1) = p^*$ sets the next item so that the predicted success probability

TABLE I
BENCHMARK COMPARISON ACROSS 10,000 SIMULATED LEARNER
TRAJECTORIES

Architecture	TTM (h)	Impasse (%)	Tokens/h	Flow (%)
Static curriculum	48.5	42.1	0	31.4
Rule-based ITS	39.2	31.8	120	44.6
LLM-only (System Two)	28.4	19.5	48,200	52.1
DP-ALS (ours)	18.6	6.2	7,400	78.9

equals the target, where $\hat{\theta}_t$ is the current ability estimate. The thresholds $\tau_{\text{flow}} = 0.70$, $\tau_P = -0.4$, and $\tau_{\text{impasse}} = 0.75$ match the values used in Fig. 2 and Appendix B. Because $\mathcal{P}_t > 0$ whenever $\mathcal{C}_t < \mathcal{C}^*$ in Eq. (5), a System Two dispatch can only occur at elevated load, as in Fig. 2.

V. EMPIRICAL EVALUATION

A. Experimental Configuration

We ran an agent-based Monte Carlo simulation of $N = 10,000$ learners in four equal cohorts of 2,500: (A) university engineering undergraduates, (B) mid-career technical reskillers, (C) non-technical upskillers, and (D) senior lifelong learners. Four architectures were benchmarked on a standard 50-module STEM curriculum: (1) a static curriculum, (2) a rule-based ITS, (3) a monolithic LLM tutor (System Two only), and (4) the proposed DP-ALS. All latency, token, and learning-outcome figures in this paper are simulation outputs, not measurements from a deployed system.

B. Results and Comparative Analysis

Fig. 2 shows an illustrative simulated session in which System One issues micro-nudges and System Two is dispatched once, and Table I and Fig. 3 summarize the aggregate results.

- **Time-to-mastery (TTM).** DP-ALS reached mastery in 18.6 h on average, versus 28.4 h for the monolithic LLM and 48.5 h for the static curriculum (a 61.6% reduction). Instant micro-nudges keep minor conceptual slips from compounding into multi-minute halts.
- **Cognitive impasse.** Impasse rates fell to 6.2%, compared with 42.1% for the static curriculum (an 85.3% reduction) and 19.5% for the monolithic LLM. We attribute the latter to latency: a multi-second wait for each LLM response degrades attention.
- **Token and energy efficiency.** By delegating 91.3% of routine pedagogical decisions to the fast System One model, token consumption fell from 48,200 to 7,400 tokens per hour, an 84.6% reduction in computational cost that supports green-computing goals.
- **Flow preservation.** Sustained flow reached 78.9%, because System One avoids intrusive interruptions while the learner is in productive struggle.

VI. ALIGNMENT WITH SDGs, KNOWLEDGE PROFILES, AND PROGRAM OUTCOMES

DP-ALS is mapped to the UN SDGs, to the Washington Accord knowledge profiles WK1–WK9, and to the program

outcomes of the IEA graduate attributes (version 4, 2021) [14], whose eleven attributes WA1–WA11 are referred to here as PO1–PO11. The full mapping is given in Table II. In brief, the state-space filtering and IRT modeling address WK2 and PO1/PO2; the System One / System Two architecture addresses WK3, WK5 and PO3; the decision, LLM, and streaming tooling addresses WK6 and PO5; the Monte Carlo study addresses PO4; the 84.6% token reduction and SDG alignment address WK5, WK7 and PO6; client-side privacy safeguards address WK9 and PO7; the compute-versus-pedagogy objective addresses PO10; and the Skill Passport addresses PO11.

VII. CONCLUSION AND FUTURE WORK

This paper presented a theoretical framework and cyber-physical realization for continuous education that moves pedagogical systems from episodic instruction to continuous, life-long adaptation. By pairing a low-latency System One decision model with a deliberate System Two Socratic LLM, DP-ALS balances responsiveness, deep personalization, and computational sustainability. Future work will address on-device neuromorphic System One deployment, federated privacy-preserving telemetry aggregation across enterprise consortia, a fairness audit of adaptive pacing across learner cohorts, and empirical trials within accredited degree programs.

APPENDIX A TELEMETRY SERIALIZATION SCHEMA

The client worker transmits binary WebSocket frames of the form

$$\text{Frame} = \langle \text{timestamp}_{\text{uint64}}, \Delta\tau_{k,\text{float32}}, H_{k,\text{float32}}, \sigma_{\text{scroll},\text{float32}}, e_{\text{syntax},\text{uint8}}, t_{\text{dwell},\text{float32}}, r_{\text{churn},\text{uint8}} \rangle.$$

The fields total 26 bytes, padded to 28 bytes per message. At one message per 50 ms this is 0.56 KB/s, enabling high-fidelity telemetry over constrained mobile links (SDG 9, PO5).

APPENDIX B SYSTEM TWO SOCRATIC CONTEXT INJECTION TEMPLATE

When an impasse is flagged ($\mathcal{P}_t < -0.4$), the prompt compiler builds structured context for the generative reasoning agent:

```
[SYSTEM CONTEXT: DUAL-PROCESS PEDAGOGICAL
AGENT]
Learner Profile: Mid-Career Professional
(Age 42)
Target Competency: Graph Algorithms
(Dijkstra, WK3/PO1)
State: C_t = 0.82 (High), Flow = 0.12,
Struggle = -0.65
Diagnostic: 3 failed attempts on relaxation
step in 45 s.
DIRECTIVE: Do NOT provide solution code.
Formulate a 2-sentence Socratic question
grounding node relaxation in logistics
routing. Preserve learner agency.
```

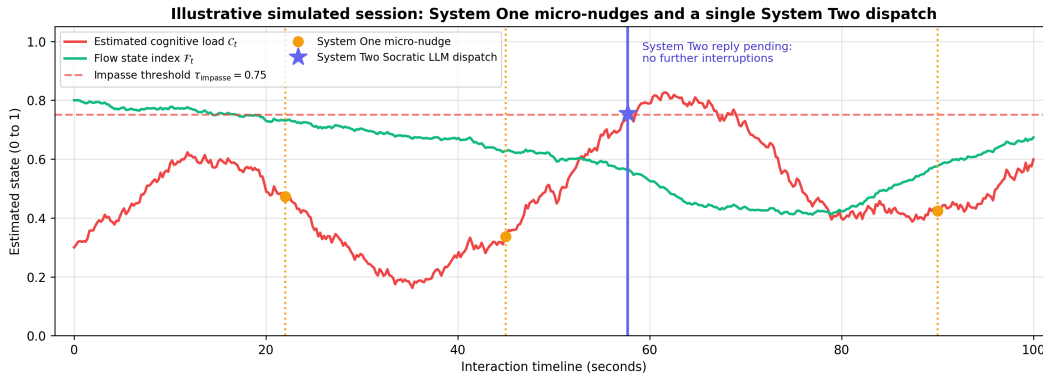


Fig. 2. Illustrative simulated session. System One tracks cognitive load C_t and flow F_t and deploys micro-nudges while load is below τ_{impasse} ; System Two is dispatched once, when the threshold is first breached.

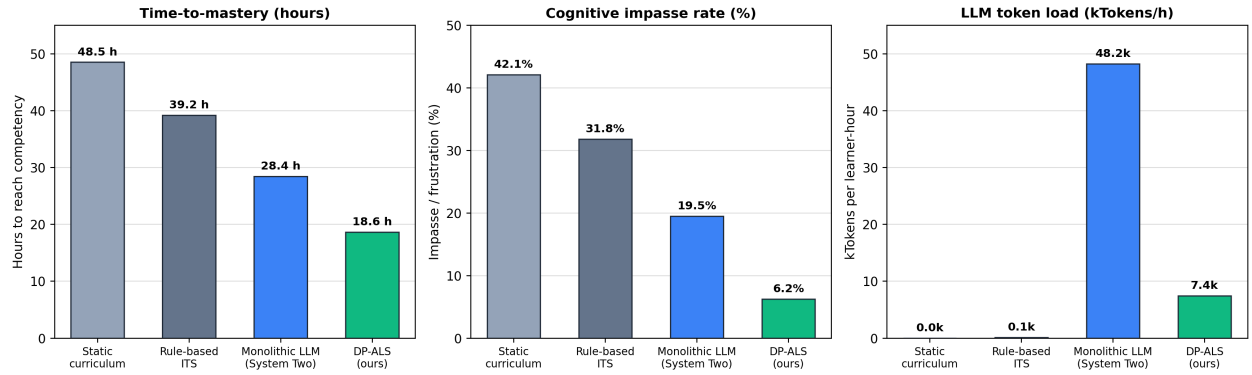


Fig. 3. Quantitative benchmarks. DP-ALS reduces time-to-mastery by 61.6% versus the static curriculum, cuts impasse to 6.2%, and reduces token consumption by 84.6% versus the monolithic LLM.

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TABLE II
ACCREDITATION AND STANDARDS ALIGNMENT OF DP-ALS (PO_n = IEA GRADUATE ATTRIBUTE WA_n [14])

Framework	ID	Descriptor	Realization in DP-ALS
UN SDGs	SDG 4	Quality and inclusive lifelong education	Adaptive scaffolding accessible to all age groups; continuous skill acquisition.
	SDG 8	Decent work and economic growth	Reskilling and upskilling modules aligned with evolving industrial demands.
	SDG 9	Industry, innovation, and infrastructure	AI educational infrastructure that optimizes compute sustainability.
	SDG 10	Reduced inequalities	Low-bandwidth telemetry (0.56 KB/s) for constrained mobile links; personalized pacing across age cohorts.
WKs	WK2	Mathematics, data analysis, and statistics	POMDP Bayesian filtering, IRT, and information-theoretic telemetry modeling.
	WK3	Engineering fundamentals	Low-latency client-server architecture, WebSockets, asynchronous pipelines.
	WK5	Engineering design knowledge, including efficient resource use	Integration of fast reflex AI, generative reasoning, and knowledge graphs; 84.6% lower LLM token consumption.
	WK6	Engineering practice (technology)	Edge inference, LLM orchestration, and binary streaming pipelines.
	WK7	Role of engineering in society; sustainable development	Societal upskilling aligned with SDGs 4, 8, 9, and 10 at a lower compute footprint.
	WK9	Ethics, inclusive behavior and conduct	Client-side privacy-preserving telemetry; adaptive pacing, with a fairness audit left to future work.
POs	PO1, PO2	Engineering knowledge; problem analysis	Formal cognitive-state estimation and impasse detection from first principles.
	PO3	Design and development of solutions	System One / System Two architecture designed around token cost, latency, and privacy constraints.
	PO4	Investigation	Agent-based Monte Carlo study of 10,000 learners with analysis of the results.
	PO5	Tool usage	Decision models, LLM orchestration, and modeling tools, with the limits of simulation stated.
	PO6	The engineer and the world	Sustainable-development impacts: SDG alignment and reduced compute.
	PO7	Ethics	GDPR/FERPA-aware design; no keystroke content captured.
	PO8	Individual and collaborative teamwork	Modular microservices enabling multidisciplinary, distributed development.
	PO10	Project management and finance	Objective that trades pedagogical return against compute cost.
	PO11	Lifelong learning	Lifelong competency tracking and micro-credential verification.